```python
def summarize_text(text: str) -> str:
    try:
        model = genai.GenerativeModel(model_name="models/gemini-1.5-pro")
        prompt = f"Summarize the following text in simple language:\n\n{text}"
        response = model.generate_content(prompt)
        return response.text.strip()
    except Exception as e:
        return f"⚠️ Error in Summary: {e}"
```

Fig. Summary Module

5. Learning Path Module

The get_learning_recommendations function uses Google Gemini to generate a personalized, structured learning path for any given topic. By crafting a detailed prompt, it instructs the AI to suggest beginner to advanced concepts, organized by difficulty, and supported with useful resources such as videos, articles, or books. It ensures adaptability to the learner's level. The function safely handles API responses and errors, providing either the generated text or a helpful error message if the response is invalid or incomplete.

```python
def get_learning_recommendations(topic):
    prompt = f"""
    You are an AI tutor. The student wants to learn about: {topic}.
    Suggest a structured and adaptive learning path including key topics, order of learning, and resources (l
    Include beginner, intermediate, and advanced levels if needed.
    """
    try:
        response = model.generate_content(prompt)
        print("🧠 Gemini raw response:", response)

        if hasattr(response, "text"):
            return response.text
        elif hasattr(response, "parts") and response.parts:
            return response.parts[0].text
        else:
            return "❌ Could not extract content from Gemini response."
    except Exception as e:
        import traceback
        traceback.print_exc()  # ✅ This gives full error trace
        return f"❌ Error occurred: {str(e)}"
```

Fig. Learning Path Module

Activity 2.2 : Backend API with FastAPI

- Defined RESTful endpoints for each module:

- /qa
- /explain
- /quiz
- /summarize
- /learn/recommendations

- Connected each endpoint to a respective module logic

main.py

```python

# Q&A - GET API using Gemini
@app.get("/qa")
async def answer_question(question: str = Query(...)):
    answer = answer_question_with_gemini(question)
    return {"answer": answer}

# Explanation - POST API
@app.post("/explain/")
async def explain_api(request: Request):
    data = await request.json()
    topic = data.get("topic")
    if not topic:
        return JSONResponse(content={"error": "Please provide a topic."}, status_code=400)
    explanation = explain_topic(topic)
    return {"topic": topic, "explanation": explanation}

# Summarization - POST API
@app.post("/summarize/")
async def summarize_api(request: Request):
    data = await request.json()
    text = data.get("text")
    if not text:
        return JSONResponse(content={"error": "Please provide text to summarize."}, status_code=400)
    summary = summarize_text(text)
    return {"summary": summary}

```

Fig. Endpoints of API

```python

# Quiz Generation - POST API
@app.post("/quiz")
async def quiz_api(request: Request):
    data = await request.json()
    text = data.get("text")
    if not text:
        return JSONResponse(content={"error": "Please provide text for quiz."}, status_code=400)
    quiz = generate_quiz(text)
    print("Generated quiz:", quiz)  # ✅ DEBUG
    return JSONResponse(content={"quiz": quiz})

# Learning Recommendations - GET API
@app.get("/learn/recommendations")
async def learning_recommendation_api(topic: str = Query(...)):
    recommendation = get_learning_recommendations(topic)
    return {"topic": topic, "recommendation": recommendation}

```

Fig. Endpoints of API

MILESTONE 3: Frontend Development

Activity 3.1: Build Web Interface

- HTML Form (index.html):
    - Task dropdown (Explain, QnA, Quiz, Summary, Recommend Path)
    - Text area for user input
    - Submit button
- CSS Styling (style.css):
    - Responsive design
    - Styled buttons, input areas, and result container

templates > index.html > ...

```
2   <html lang="en">
3   <head>
86  </head>
87  <body>
88      <h1>Welcome to EduGenie 🧠✨</h1>
89      <p>Your personal AI tutor for learning support!</p>
90
91      <!-- Q&A -->
92      <form id="qaForm">
93          <label><strong>Ask EduGenie a Question:</strong></label><br><br>
94          <input type="text" id="question" placeholder="Why is the sky blue?" required>
95          <button type="submit">Get Answer</button>
96      </form>
97      <div id="qaResult" class="output"></div>
98
99      <!-- Explanation -->
100     <form id="explainForm">
101         <label><strong>Need an Explanation?</strong></label><br><br>
102         <input type="text" id="topic" placeholder="Photosynthesis" required>
103         <button type="submit">Explain</button>
104     </form>
105     <div id="explanationResult" class="output"></div>
106
107     <!-- Summary -->
108     <form id="summaryForm">
109         <label><strong>Summarize a Paragraph:</strong></label><br><br>
110         <input type="text" id="summaryText" placeholder="Paste long content to summarize" required>
111         <button type="submit">Summarize</button>
112     </form>
113     <div id="summaryResult" class="output"></div>
114
115     <!-- Quiz -->
116     <form id="quizForm">
117         <label><strong>Generate a Quiz:</strong></label><br><br>
118         <input type="text" id="quizText" placeholder="Solar System" required>
119         <button type="submit">Generate Quiz</button>
120     </form>
121     <div id="quizResult" class="output"></div>
```